

# Set Topology

## Lecture 05: Topological Spaces

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### Topics Covered

- |                        |                     |
|------------------------|---------------------|
| ✓ Topological Spaces   | ✓ Metric Topology   |
| ✓ Trivial & Discrete   | ✓ Cofinite Topology |
| ✓ Cocountable Topology | ✓ Finer / Coarser   |

*Based on Introduction to Topological Manifolds by John M. Lee (2nd ed.) Point-Set Topology Series | Undergraduate / Early Graduate*

# 1. Introduction and Motivation

💡 *Think of topology as geometry without a ruler.*

**Topology** is the branch of mathematics concerned with the properties of space that are preserved under continuous deformations — stretching, twisting, and bending — but *not* tearing, gluing, or puncturing. It is sometimes called *rubber-sheet geometry*.

## 1.1. Prerequisites at a Glance

### ✓ Before You Begin

- |                                  |                                                  |
|----------------------------------|--------------------------------------------------|
| ✓ Sets and functions             | ✓ $\varepsilon$ - $\delta$ convergence           |
| ✓ Euclidean space $\mathbb{R}^n$ | ● Open sets in $\mathbb{R}^n$ ( <i>helpful</i> ) |
| ✓ Continuous functions           | ● Sequences in $\mathbb{R}^n$ ( <i>helpful</i> ) |

## 1.2. The Guiding Question

The starting point is deceptively simple:

*“When is one space essentially the same as another?”*

To answer this we need a language precise enough to distinguish a doughnut from a sphere, yet flexible enough to work on spaces with no notion of distance.

# 2. Motivating Examples

⚠ *These three spaces look alike locally but differ globally.*

## 2.1. Euclidean Space $\mathbb{R}^3$ — Where Topology Begins

In calculus we say a sequence  $(x_n) \subset \mathbb{R}^3$  **converges** to  $x$  if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ s.t. } n \geq N \implies d(x_n, x) < \varepsilon,$$

where  $d$  is the Euclidean distance. Equivalently, every open ball  $B_\varepsilon(x) = \{y \in \mathbb{R}^3 : d(x, y) < \varepsilon\}$  eventually contains all terms of the sequence.

### 💡 Key Insight — From $\varepsilon$ -balls to Open Sets

The ball  $B_\varepsilon(x)$  is itself an **open set**. Replacing “ball” by “open set” in the definition of convergence gives a statement that makes sense on *any topological space* — no distances required. This is the fundamental philosophical shift from analysis to topology.

## 2.2. The 2-Sphere $S^2$ vs. The Plane $\mathbb{R}^2$

### 📖 Definition: The 2-Sphere

$S^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\}$ , the surface of the unit ball.

| Locally (zoom in)                                                                                        | Globally (step back)                                                                       |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Both $S^2$ and $\mathbb{R}^2$ look flat — an ant on a large sphere cannot tell the surface from a plane. | Walk straight on $S^2$ and you return to your start. On $\mathbb{R}^2$ you travel forever. |

### 2.3. The Sphere $S^2$ vs. the Torus $T^2$

💡 *Holes are the key topological invariant here.*

The torus  $T^2$  is the surface of a doughnut. Both  $S^2$  and  $T^2$  let you “walk in a circle”, but they differ in a subtler way:

#### Loop Contraction Test

- **On  $S^2$ :** Every closed loop can be continuously contracted to a single point.
- **On  $T^2$ :** A loop that goes *through the hole* cannot be shrunk to a point. There are two independent non-contractible loops: one through the hole, one around it.

This distinction is captured rigorously by the *fundamental group*:  $\pi_1(S^2) = 0$  (trivial), while  $\pi_1(T^2) \cong \mathbb{Z} \times \mathbb{Z}$ . We will build the machinery to prove this in the algebraic topology portion of the course.

## 3. The Formal Definition of a Topological Space

★ *This is the single most important definition in the course.*

#### 📖 Definition: Topology on a Set $X$

Let  $X$  be a non-empty set. A **topology** on  $X$  is a collection  $\mathcal{T}$  of subsets of  $X$  satisfying:

**(T1)**  $\emptyset \in \mathcal{T}$  and  $X \in \mathcal{T}$ .

**(T2) (Finite intersections)** If  $U_1, U_2, \dots, U_n \in \mathcal{T}$  for some  $n \in \mathbb{N}$ , then  $\bigcap_{i=1}^n U_i \in \mathcal{T}$ .

**(T3) (Arbitrary unions)** If  $\{U_\alpha\}_{\alpha \in \Lambda}$  is *any* collection of elements of  $\mathcal{T}$  (possibly infinite), then  $\bigcup_{\alpha \in \Lambda} U_\alpha \in \mathcal{T}$ .

The pair  $(X, \mathcal{T})$  is called a **topological space**. Elements of  $\mathcal{T}$  are called the **open sets** of  $X$ .

### 3.1. Why Finite vs. Arbitrary?

#### ⚠ Common Mistake / Caution

**Axiom (T2) is finite-only on purpose!** Infinite intersections of open sets need *not* be open.

**Counterexample in  $\mathbb{R}$ :** Each open interval  $(-\frac{1}{n}, \frac{1}{n})$  is open, yet

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\},$$

which is a single point — *not* open in  $\mathbb{R}$ .

**Axiom (T3), by contrast, allows infinite unions.** Topology is more “generous” with unions than with intersections.

### 3.2. Open Sets as Neighbourhoods

💡 *Topologies encode nearness without distance.*

Each open set  $U \ni x$  acts as a *neighbourhood* of  $x$ : a region surrounding  $x$  within which we know  $x$  lives. The axioms then say:

- The whole space and the empty set are always neighbourhoods (T1).
- The overlap of finitely many neighbourhoods of  $x$  is a neighbourhood of  $x$  (T2).
- Any union of open sets — no matter how many — is open (T3).

## 4. Four Fundamental Topologies

### 4.1. The Trivial (Indiscrete) Topology

#### 📖 Definition: Trivial Topology

For any non-empty set  $X$ , define  $\mathcal{T}_0 = \{\emptyset, X\}$ .

#### Proof

**Verification of axioms.**

- (T1)  $\emptyset \in \mathcal{T}_0$  and  $X \in \mathcal{T}_0$  by definition. ✓
- (T2) The only non-empty element is  $X$ . Any intersection involving  $\emptyset$  gives  $\emptyset$ ; otherwise all factors equal  $X$  so the intersection is  $X$ . ✓
- (T3) Any union containing  $X$  gives  $X$ ; a union of copies of  $\emptyset$  gives  $\emptyset$ . ✓

#### ★ Why “Coarsest”?

$\mathcal{T}_0 = \{\emptyset, X\}$  is the **coarsest** (smallest) topology on  $X$ : it is contained in every other topology. Any topology  $\mathcal{T}$  on  $X$  must satisfy (T1), so it must contain  $\emptyset$  and  $X$  — hence  $\mathcal{T}_0 \subseteq \mathcal{T}$  always holds.

## 4.2. The Discrete Topology

### Definition: Discrete Topology

For any non-empty set  $X$ , define  $\mathcal{T}_1 = \mathcal{P}(X)$ , the **power set** of  $X$  (the set of all subsets of  $X$ ).

### Proof

**Verification.** Any union or intersection of subsets of  $X$  is again a subset of  $X$ , hence in  $\mathcal{P}(X) = \mathcal{T}_1$ . All three axioms hold trivially. ✓

### ★ Why “Finest”?

$\mathcal{T}_1 = \mathcal{P}(X)$  is the **finest** (largest) topology on  $X$ : every topology  $\mathcal{T}$  satisfies  $\mathcal{T} \subseteq \mathcal{P}(X) = \mathcal{T}_1$ .

### Key Insight — Discrete = Isolated Points

In the discrete topology, every singleton  $\{x\}$  is open. Intuitively, every point is “isolated” from every other — there is no notion of being “close to” any other point. This is as far from geometry as you can get while still having a topology.

## 4.3. A Concrete Finite Example

### Example: Nested Topology on a 3-Element Set

Let  $Z = \{1, 2, 3\}$  and

$$\mathcal{T} = \{ \emptyset, \{1\}, \{1, 2\}, \{1, 2, 3\} \}.$$

**Observation:** The open sets form a *chain* under inclusion:

$$\emptyset \subset \{1\} \subset \{1, 2\} \subset \{1, 2, 3\}.$$

**Verification.**

(T1)  $\emptyset, Z \in \mathcal{T}$ . ✓

(T2) Any intersection of chain members equals the smallest member of the chosen subcollection — which is in  $\mathcal{T}$ . E.g.  $\{1\} \cap \{1, 2\} = \{1\} \in \mathcal{T}$ . ✓

(T3) Any union equals the largest member — also in  $\mathcal{T}$ . E.g.  $\{1\} \cup \{1, 2\} = \{1, 2\} \in \mathcal{T}$ . ✓

Note that  $\{2\}$ ,  $\{3\}$ ,  $\{2, 3\}$ , and  $\{1, 3\}$  are *not* open sets in this topology.

#### 4.4. The Metric Topology

##### Definition: Metric Space

A **metric space**  $(M, d)$  consists of a set  $M$  and a function  $d : M \times M \rightarrow \mathbb{R}_{\geq 0}$  satisfying:

- (i)  $d(x, y) = 0 \iff x = y$  (identity of indiscernibles),
- (ii)  $d(x, y) = d(y, x)$  (symmetry),
- (iii)  $d(x, z) \leq d(x, y) + d(y, z)$  (triangle inequality).

##### Definition: Open Ball and Metric Topology

Given  $(M, d)$ , the **open ball** of radius  $\varepsilon > 0$  centred at  $x$  is

$$B_\varepsilon(x) = \{y \in M : d(x, y) < \varepsilon\}.$$

A subset  $U \subseteq M$  is **open** (in the metric sense) if

$$\forall x \in U \exists \varepsilon > 0 \text{ s.t. } B_\varepsilon(x) \subseteq U.$$

The collection of all such open sets forms the **metric topology** on  $M$ .

★ Every metric space carries a natural topology.

##### Example: Open Sets in $\mathbb{R}$

Take  $M = \mathbb{R}$ ,  $d(x, y) = |x - y|$ . Then  $B_\varepsilon(x) = (x - \varepsilon, x + \varepsilon)$ .

| Set                  | Open? | Reason                                                           |
|----------------------|-------|------------------------------------------------------------------|
| $(a, b)$             | ✓ Yes | Every interior point has a ball inside.                          |
| $[a, b]$             | ✗ No  | Endpoints have no ball inside $[a, b]$ .                         |
| $\mathbb{R}$         | ✓ Yes | Take any $\varepsilon$ : $B_\varepsilon(x) \subset \mathbb{R}$ . |
| $\emptyset$          | ✓ Yes | Vacuously true.                                                  |
| $\{p\}$ (point)      | ✗ No  | $B_\varepsilon(p)$ always contains other points.                 |
| $(0, 1) \cup (2, 3)$ | ✓ Yes | Union of two open sets.                                          |

#### 5. The Cofinite and Cocountable Topologies

💡 These topologies arise from complements, not from distance.

## 5.1. Cofinite Topology

### Definition: Cofinite Topology $\mathcal{T}_f$

Let  $X$  be any non-empty set. Define

$$\mathcal{T}_f = \{ U \subseteq X : X \setminus U \text{ is finite} \} \cup \{ \emptyset \}.$$

Equivalently,  $U$  is open if its complement  $U^c = X \setminus U$  is *finite* or equal to all of  $X$ .

### Proof

**Verification of all three axioms.**

**(T1)**  $X \setminus X = \emptyset$  is finite, so  $X \in \mathcal{T}_f$ .  $\emptyset$  is included directly. ✓

**(T2)** Let  $U_1, U_2 \in \mathcal{T}_f$  with  $U_1, U_2 \neq \emptyset$ . Then  $U_1^c$  and  $U_2^c$  are finite. By De Morgan's law:

$$(U_1 \cap U_2)^c = U_1^c \cup U_2^c.$$

A union of two finite sets is finite, so  $U_1 \cap U_2 \in \mathcal{T}_f$ . ✓

**(T3)** Let  $\{U_\alpha\} \subseteq \mathcal{T}_f$  be any non-empty collection. Pick any  $U_{\alpha_0} \neq \emptyset$ . Then by De Morgan:

$$\left( \bigcup_{\alpha} U_{\alpha} \right)^c = \bigcap_{\alpha} U_{\alpha}^c \subseteq U_{\alpha_0}^c,$$

which is finite. Hence  $\bigcup U_{\alpha} \in \mathcal{T}_f$ . ✓

### Example: Cofinite Topology on $\mathbb{R}$

In  $(\mathbb{R}, \mathcal{T}_f)$ :

- $\mathbb{R} \setminus \{1, 2, 3\}$  is open (complement is finite).
- $(0, 1)$  is *not* open — its complement  $(-\infty, 0] \cup [1, \infty)$  is infinite.
- Any finite subset of  $\mathbb{R}$  is *closed* (its complement is cofinite, hence open).

**Comparison with the standard topology on  $\mathbb{R}$ :**  $\mathcal{T}_{\text{std}}$  is strictly finer than  $\mathcal{T}_f$  — every cofinite open set is also open in the standard topology, but not vice versa.

## 5.2. Cocountable Topology

### Definition: Cocountable Topology $\mathcal{T}_c$

For a non-empty set  $X$ , define

$$\mathcal{T}_c = \{ U \subseteq X : X \setminus U \text{ is countable} \} \cup \{ \emptyset \}.$$

The proof is entirely analogous to the cofinite case: replace *finite* with *countable* everywhere, and note that countable unions of countable sets are countable.

### ★ Hierarchy on an Uncountable Set

When  $X$  is uncountable (e.g.  $X = \mathbb{R}$ ), we have the strict chain:

$$\mathcal{T}_0 \subsetneq \mathcal{T}_c \subsetneq \mathcal{T}_f \subsetneq \mathcal{T}_{\text{std}} \subsetneq \mathcal{T}_1.$$

|                         |           |                                     |           |                          |
|-------------------------|-----------|-------------------------------------|-----------|--------------------------|
| Trivial $\mathcal{T}_0$ | $\subset$ | Cocountable $\mathcal{T}_c$         | $\subset$ | Cofinite $\mathcal{T}_f$ |
|                         |           | $\subset$                           |           | Discrete $\mathcal{T}_1$ |
|                         |           | Standard $\mathcal{T}_{\text{std}}$ |           |                          |

$\longleftarrow$  coarser
 $\longrightarrow$  finer

## 6. Comparing Topologies: Finer and Coarser

### 📖 Definition: Finer and Coarser Topologies

Let  $\mathcal{T}_1$  and  $\mathcal{T}_2$  be topologies on the same set  $X$ .

- $\mathcal{T}_1$  is **finer** than  $\mathcal{T}_2$  (or  $\mathcal{T}_2$  is **coarser** than  $\mathcal{T}_1$ ) if  $\mathcal{T}_2 \subseteq \mathcal{T}_1$ .
- They are **comparable** if one contains the other; otherwise **incomparable**.

💡 “Finer” = more open sets = more structure.

### ✍ Example: Incomparable Topologies

Let  $X = \{a, b, c\}$  and consider:

$$\mathcal{T}_A = \{\emptyset, \{a\}, X\}, \quad \mathcal{T}_B = \{\emptyset, \{b\}, X\}.$$

Neither  $\mathcal{T}_A \subseteq \mathcal{T}_B$  (since  $\{a\} \notin \mathcal{T}_B$ ) nor  $\mathcal{T}_B \subseteq \mathcal{T}_A$  (since  $\{b\} \notin \mathcal{T}_A$ ). They are **incomparable**.

### ★ Intersection of Topologies is a Topology

If  $\{\mathcal{T}_\alpha\}$  is any collection of topologies on  $X$ , then  $\mathcal{T} = \bigcap_\alpha \mathcal{T}_\alpha$  is also a topology on  $X$ .

**Proof sketch:** Each axiom passes through arbitrary intersections: a set belongs to  $\mathcal{T}$  iff it belongs to every  $\mathcal{T}_\alpha$ , and each  $\mathcal{T}_\alpha$  is closed under finite intersections and arbitrary unions.  $\square$

## 7. Summary Table



| Topology                           | Open Sets                           | Notes                   |
|------------------------------------|-------------------------------------|-------------------------|
| <b>Trivial</b> $\mathcal{T}_0$     | $\{\emptyset, X\}$ only             | Coarsest possible       |
| <b>Discrete</b> $\mathcal{T}_1$    | All subsets of $X$                  | Finest possible         |
| <b>Metric topology</b>             | Sets with interior-point property   | Generalises analysis    |
| <b>Cofinite</b> $\mathcal{T}_f$    | Complements are finite (or $X$ )    | Useful on infinite sets |
| <b>Cocountable</b> $\mathcal{T}_c$ | Complements are countable (or $X$ ) | Coarser than cofinite   |

## 8. Practice Problems

### Practice Problems

**P1. (Verification)** Let  $X = \{a, b, c\}$ . For each collection below, determine whether it is a topology. If not, state which axiom fails and give the *minimal fix*.

(a)  $\mathcal{T}_1 = \{\emptyset, X, \{a\}, \{b\}\}$

*Hint: check unions.*

(b)  $\mathcal{T}_2 = \{\emptyset, X, \{a, b\}, \{b, c\}\}$

*Hint: check intersections.*

(c)  $\mathcal{T}_3 = \{\emptyset, X, \{a\}, \{b, c\}\}$

**P2. (Complete enumeration)** Find *all* distinct topologies on  $X = \{a, b\}$ . Which is finest? Which is coarsest? Are any incomparable?

**P3. (Union of topologies)** Let  $\mathcal{T}_1$  and  $\mathcal{T}_2$  be topologies on  $X$ .

(a) Prove that  $\mathcal{T}_1 \cap \mathcal{T}_2$  is a topology on  $X$ .

(b) Give an example showing that  $\mathcal{T}_1 \cup \mathcal{T}_2$  need *not* be a topology. *Hint: try  $X = \{1, 2, 3\}$ .*

**P4. (Cofinite vs. standard on  $\mathbb{R}$ )**

(a) Show that  $\mathcal{T}_f \subsetneq \mathcal{T}_{\text{std}}$  (strictly coarser).

*Hint: use De Morgan.*

(b) Is  $\mathcal{T}_c \subsetneq \mathcal{T}_f$ ? Prove or disprove.

**P5. (Metric topology)** Let  $(M, d) = (\mathbb{R}^2, d_\infty)$  where  $d_\infty((x_1, y_1), (x_2, y_2)) = \max(|x_1 - x_2|, |y_1 - y_2|)$ .

(a) Describe the open balls  $B_\varepsilon(p)$  geometrically.

*Hint: squares, not discs.*

(b) Show that the metric topology induced by  $d_\infty$  equals the standard topology on  $\mathbb{R}^2$ .

## 9. References and Further Reading

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### Recommended Resources

- **[Primary]** Lee, J.M. *Introduction to Topological Manifolds* (2nd ed., 2011). Springer GTM 202. — §2.1–2.3.
- Munkres, J. R. *Topology* (2nd ed., 2000). Prentice Hall. — Chapter 2.
- Hatcher, A. *Algebraic Topology* (2002). Cambridge University Press. Free at <https://math.cornell.edu/~hatcher/AT/ATpage.html>
- Armstrong, M. A. *Basic Topology* (1983). Springer. — Very accessible with many pictures.
- MIT OpenCourseWare 18.901 — General Topology. <https://ocw.mit.edu>

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*Next Lecture → Closed Sets, Closure, Interior, and Boundary*

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